

1.5 Chemical Energetics

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.5 Chemical Energetics
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Which quantity gives the best indication of the relative strengths of the hydrogen bonds between water molecules in the liquid state?

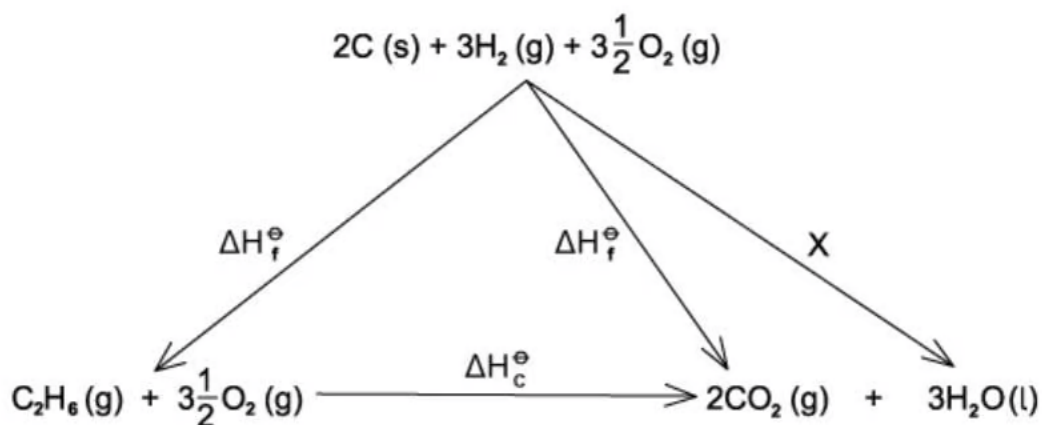
- A. Enthalpy changes of vaporisation
- B. Bond dissociation energies
- C. Enthalpy of formation
- D. Activation energy

[1 mark]

Question 2

Enthalpy changes that are difficult to measure directly can often be determined using Hess' Law to construct an enthalpy cycle.

Which enthalpy change is indicated by X in the enthalpy cycle shown?



- A. +1 x Enthalpy of formation of water
- B. -1 x Enthalpy of formation of water
- C. +3 x Enthalpy of formation of water
- D. -3 x Enthalpy of formation of water

[1 mark]

Question 3

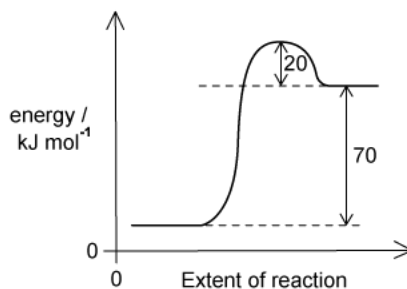
Which equation below can represent both an enthalpy change of formation and combustion?

- A. $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$
- B. $4\text{Na}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{Na}_2\text{O}(\text{s})$
- C. $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- D. $\text{CO}(\text{g}) + \text{C}(\text{s}) \rightarrow \text{CO}_2(\text{g})$

[1 mark]

Question 4

The reaction pathway for a reversible reaction is shown below:



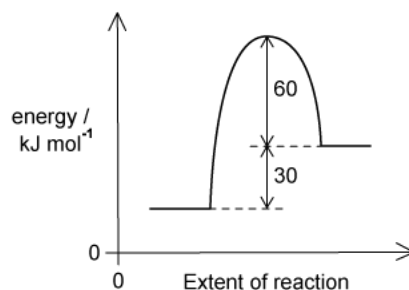
Which statement is correct?

- A. The activation energy of the reverse reaction is $+90 \text{ kJ mol}^{-1}$
- B. The activation energy of the forward reaction is $+20 \text{ kJ mol}^{-1}$
- C. The activation of the reverse reaction is $+20 \text{ kJ mol}^{-1}$
- D. The enthalpy change of forwards reaction is -70 kJ mol^{-1}

[1 mark]

Question 5

The reaction pathway for a reversible reaction is shown below.



Which statements are correct?

1	The forward reaction is endothermic
2	The enthalpy change for the backward reaction is -30 kJ mol^{-1}
3	The activation energy for the forward reaction is $+90 \text{ kJ mol}^{-1}$

A. 1 only

B. 1 and 2

C. 2 and 3

D. 1, 2 and 3

[1 mark]

Question 6

Which statement best describes why a reaction is said to be thermodynamically possible yet kinetically controlled?

A. The enthalpy of the reactants is higher than the products, and the reaction is very slow.

B. The enthalpy of the reactants is lower than the products, and the reaction is fast.

C. The enthalpy of the reactants is lower than the products, and the reaction is very slow.

D. The reaction is exothermic and very fast.

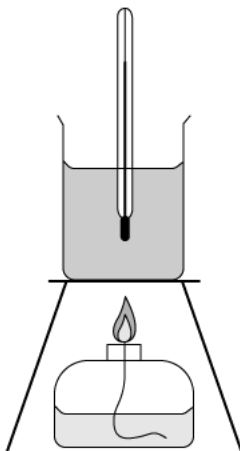
[1 mark]

Question 7

Use of the Periodic table is relevant to this question.

A student carried out an experiment to determine the enthalpy change for the combustion of ethanol.

The following results were obtained by the student. The specific heat capacity of water is $4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$.



Start temperature of the water	21 °C
Final temperature of the water	54 °C
Mass of alcohol burner before burning	259.75 g
Mass of alcohol burner after burning	259.18 g
Mass of glass beaker plus water	150.00 g
Mass of glass beaker	50.0 g

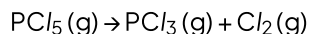
How much of the heat energy produced by the burning of ethanol went into the water?

- A. 6897 J
- B. 20691 J
- C. 22572 J
- D. 13794 J

[1 mark]

Question 8

$\text{PCl}_5(\text{g})$ dissociates as follows.



The yield of the products increases as the temperature is raised from 80°C to 110°C .

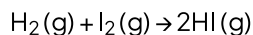
Which row correctly describes the formation of $\text{PCl}_3(\text{g})$?

	Shape of PCl_3 molecule	The reaction is
A	Pyramidal	Endothermic
B	Pyramidal	Exothermic
C	Trigonal	Endothermic
D	Trigonal	Exothermic

[1 mark]

Question 9

Hydrogen atoms bond covalently to Iodine atoms to form hydrogen iodide as shown in the equation below:



Which statement best describes what is meant by the average HI bond enthalpy?

- A. The energy stored in a covalent bond.
- B. The energy required to break one covalent bond in the gas phase.
- C. The energy required to break one mole of the HI bonds in the gas phase.
- D. The energy released when two atoms form a covalent bond.

[1 mark]

Question 10

The enthalpy of atomisation of a compound can be calculated using a range of different enthalpy changes.

Which statement below correctly describes the enthalpy change of atomisation?

- A. The energy stored in a covalent bond.
- B. The energy required to break one covalent bond in the gas phase.
- C. The energy required to break all bonds in an element.
- D. The energy released when two atoms form a covalent bond.

[1 mark]

Model Answers: Easy

1

The correct answer is **A** because:

- **Vaporisation** is the process by which a liquid is converted into a gas. Water molecules are held together in the liquid state due to the hydrogen bonds between the molecules (1 of 3 types of intermolecular forces).
- Vaporisation is achieved by **overcoming the hydrogen bonds**. The amount of energy required to overcome these forces would, therefore, be proportional to the strength of the hydrogen bond.
- Think of boiling the kettle. Steam is produced, not hydrogen and oxygen. Remember, **the covalent** bonds between the atoms are **not being broken**.

B is incorrect as this is the energy associated with the breaking of a covalent bond, not an intermolecular force.

C is incorrect as the enthalpy change of formation is defined as the enthalpy change when one mole of a substance is formed from its elements in their standard states. This, therefore, does not apply to the breaking of intermolecular forces.

D is incorrect as activation energy is the minimum energy required for a chemical reaction to occur. Vaporisation is a physical process, not a chemical reaction.

2

The correct answer is **C** because:

- The enthalpy of formation is defined as the enthalpy change when 1 **mole** of a compound is formed from its elements in their standard states and under standard conditions.
- X does show $\text{H}_2\text{O}(\text{l})$ being formed from $\text{H}_2(\text{g})$ and $\text{O}_2(\text{g})$ however, **3 moles** is produced hence $\times 3$ enthalpy of formation of water.

- Using the Hess cycle, we are travelling in the **same direction** as the arrows in our alternative route (via the enthalpy of formation then combustion of C_2H_6 and water), therefore it is **positive**.

3

The correct answer is **B** because:

- The enthalpy of formation is defined as the enthalpy change when **1 mole of a compound is formed from its elements** in their standard states and under standard conditions.
- We can see that 1 mole of Na_2O is formed from the elements Na (s) and $\text{O}_2 \text{ (g)}$. Both of which are in their standard states, therefore this is an enthalpy change of formation.
- The enthalpy of combustion is defined as the enthalpy change when **1 mole of a substance reacts completely with oxygen** under standard conditions.
- 1 mole of a substance** Na (s) has reacted completely with $\text{O}_2 \text{ (g)}$ as the oxide of sodium, Na_2O , has been produced. No more any more oxygen atoms can be added to this compound. This equation, therefore, agrees with the definition of enthalpy change of combustion.

A is incorrect as this is not enthalpy of formation as there are two compounds produced and the starting materials are not elements.

C is incorrect as this reaction does not show either formation or combustion because 1) the reactants aren't in their standard states and 2) for a reaction to be classed as combustion, **oxygen** must be added to a reactant in order to undergo combustion (burning).

D is incorrect as for a reaction to be classed as combustion, **oxygen** must be added to a reactant in order to undergo combustion (burning).

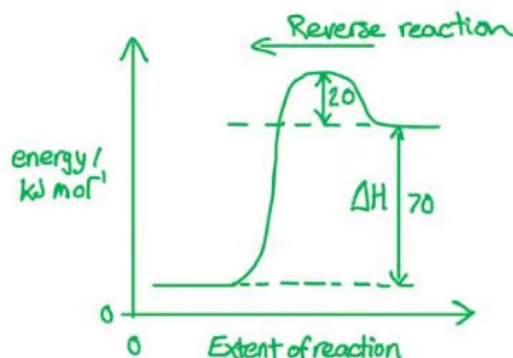
4

The correct answer is **C** because:

- The activation energy can be measured on a reaction pathway by measuring the

energy difference from the energy level to the top of the 'hump'.

- **If you look at the reversible reaction** The reaction is going from right to left here. Taking the measurement from the energy level on the right to the top of the hump which is $+20 \text{ kJ mol}^{-1}$.



A is incorrect as this is the activation energy plus the enthalpy change (the difference between the two energy levels) see diagram.

B is incorrect as to measure the activation energy, you must measure the difference from the energy level to the top of the hump. Going in the forwards direction you would, therefore, do $70 \text{ kJ mol}^{-1} + 20 \text{ kJ mol}^{-1} = +90 \text{ kJ mol}^{-1}$.

D is incorrect as the forwards reaction is endothermic and hence must have a positive sign.

5

The correct answer is **D** because:

- The reactants have less energy than the products, so energy is taken into the reaction and therefore endothermic.
- The enthalpy change of the backwards reaction (going from products to reactants) is equal to -30 kJ mol^{-1} .
- The activation energy for the forward reaction is the total energy needed to go from reactants to products, and in this case it is $+90 \text{ kJ mol}^{-1}$.

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6

The correct answer is **A** because:

- A is exothermic. Exothermic reactions are described as thermodynamically possible as the reaction is proceeding from high energy reactants to lower-energy products. This means the reaction is moving towards greater thermodynamic stability.
- 'Kinetics' means the speed of a reaction. To control the speed means the reaction is slow. Another way of explaining this is to say, 'kinetically controlled'.

B is incorrect as this reaction is endothermic. Only **some** endothermic reactions are thermodynamically possible. Remember the question is asking which statement 'best describes' being thermodynamically possible. It also has a fast rate of reaction. This means, kinetically, it is not controlled.

C is incorrect as this reaction is endothermic.

D is incorrect as whilst this reaction is thermodynamically possible (as it is exothermic) it is not kinetically controlled as it is very fast.

7

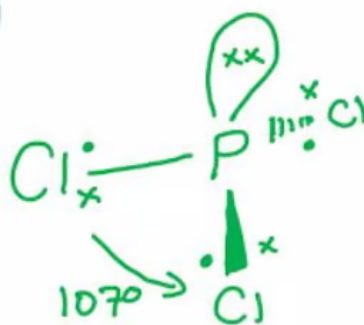
The correct answer is **D** because:

- **The energy transferred as heat is given by the equation; $q = mc\Delta T$**
 - Where q is heat transferred (J), m is the mass of water (g), c is the specific heat capacity ($\text{J g}^{-1} \text{ } ^\circ\text{C}^{-1}$), ΔT is the temperature change ($^\circ\text{C}$).
- **Known quantities:**
 - Mass of water; $150 - 50 = 100 \text{ g}$
 - Temperature change; $54 - 21 = 33 \text{ } ^\circ\text{C}$
 - Specific heat capacity of water; $4.18 \text{ J g}^{-1} \text{ } ^\circ\text{C}^{-1}$
- **Substitution into the equation:**
 - $q = mc\Delta T = 100 \times 4.18 \times 33 = 13794 \text{ J}$

8

The correct answer is **A** because:

- As more heat energy is applied to $\text{PCl}_5 (\text{g})$ the extent of dissociation has increased. This means the reaction must be endothermic.
- The valence shell of the phosphorous atom in $\text{PCl}_3 (\text{g})$ contains 8 electrons arranged in 3 bonded pairs with the 3 chlorine atoms and 1 lone pair from the phosphorous atom (phosphorous is in group 5).
- There is equal repulsion between bonding pairs but greater repulsion from the lone pair (2.5°) to the bonded pairs. Hence a pyramidal shape.



B is incorrect as the reaction is endothermic not exothermic.

C is incorrect as a trigonal shape is only possible when there are 3 bonded pairs and no lone pairs.

D is incorrect as the reaction is endothermic, not exothermic. A trigonal shape is only possible when there are 3 bonded pairs and no lone pairs.

9

The correct answer is **C** because:

- This is the exact definition of the average bond enthalpy and is applied specifically to the HI (g) bond, which is what the question asked.
- Key points from the definition to remember are:
 - The molecule must be in **the gas phase**.
 - Has to apply to **one mole**.
 - **Always endothermic** as it's the 'energy required'.

A is incorrect as this is the wrong terminology. Whilst you can picture in your mind the energy is stored, it's important to remember that the definition is very specific. '**Energy required to break..**' are the words to use. It is also an incomplete definition. Fails to touch on all points listed.

B is incorrect as the definition is for 1 mole of bonds rather than just a unitless number. You must learn the definition word for word..

D is incorrect as the average bond enthalpy relates to **energy required not released**.

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D is incorrect as the average bond enthalpy relates to **energy required not released**.

10

The correct answer is **C** because:

- The enthalpy of atomisation ($\Delta H_{at}^\ominus$) is the enthalpy change when 1 mole of gaseous atoms is formed from its elements under standard conditions.
- Key points from the definition to remember are:
 - The molecule must be in **the gas phase**.
 - **One mole**.
 - **Always endothermic** as it's the 'energy required'.
- All the bonds holding the atoms together need to be broken.

A is incorrect as this is the wrong terminology. Whilst you can picture in your mind the energy is stored, it's important to remember that the definition is very specific.

B is incorrect as the definition is for 1 mole of bonds rather than just a unitless number. You must learn the definition word for word.

D is incorrect as energy is supplied to the reaction to break the bonds holding the atoms together.